Type-Aware Failure Prediction for Chart Reasoning in Vision-Language Models

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Abstract

A vision-language model that answers a chart question wrongly may have misread the figure or stated something the figure does not show. We ask whether these failure types are distinguishable in the model's internal state before it answers, and whether the type should decide the repair. On ChartQA, a linear probe on the last prompt token of Qwen2.5-VL-7B and LLaVA-NeXT-7B predicts wrong answers with AUROC 0.88 and 0.80, well above surface features of the question. Fabrication is almost absent from real chart errors. The real errors are misreads and arithmetic errors, which probes separate (0.90 and 0.96), also within a single question type. On synthetic bar charts that ask about missing categories, neither model refuses, yet both encode the absence (AUROC 0.96 to 1.00), and the same signal flags years outside a chart's range. In a pre-registered repair experiment, cropping to the asked bar rescues misreads more than fabrications in both models. A type-aware controller beats binary selective prediction only for Qwen, mainly by turning missing-bar answers into refusals. Whether a chart can answer the question is the most useful internal signal for abstention, while misread signals on synthetic charts largely reflect chart geometry.

1 Introduction

Vision-language models (VLMs) are routinely asked to read charts, and a wrong answer can be wrong in two different ways. The model may have *misread* the figure: taken the adjacent bar, rounded to a gridline, or misread a printed value. The answer was in the image, and a closer look might recover it. Or it may have *fabricated*: reported a value the figure does not contain. Looking again cannot help, and the right response is to withhold the answer. Reliability methods usually collapse both cases into one "do not trust" score. We ask whether the two failure types are visible in the model's internal state *before* it answers, and whether knowing the type helps choose a repair.

Problem. Let $x = (I, q)$ be a chart image and a question, and $\hat{y}$ the answer a frozen VLM generates greedily. Let $h_\ell(x) \in \mathbb{R}^d$ be the hidden state after decoder layer ℓ at the last prompt token, computed in the same forward pass that produces $\hat{y}$. Each wrong answer has a failure type t, read off $\hat{y}$ and the figure. We test (i) whether linear probes on h_ℓ predict $\Pr(\hat{y}$ wrong$)$ and t better than surface features of the input, (ii) whether a probe trained for one type also responds to the other (cross-transfer), (iii) whether the best repair depends on t (E4), and (iv) whether a controller that uses the type beats plain selective prediction (E5). The **Type Separability Hypothesis** states that the states before a misread and before a fabrication are linearly distinguishable, the first being a precision-limited perceptual failure and the second an answer ungrounded in the image. A collapse of the two classes is a reportable outcome, not a failure of the design.

Contributions. (1) Full-depth activation caches of Qwen2.5-VL-7B and LLaVA-NeXT-7B on ChartQA and on three synthetic pilots, taken from the pass that produced each answer.[1] (2) Five-label annotations of all 1,495 ChartQA errors of both models, with a two-person agreement check. (3) Probes that predict wrong answers (AUROC 0.877 and 0.798) and separate misreads from arithmetic errors on real charts (0.897 and 0.961), each against surface baselines, residualisation, five data splits and single-question-type controls. (4) A synthetic arm that induces fabrication: neither model refuses questions about a missing bar, yet both encode that it is missing (AUROC 0.96 to 1.00), and the same signal fires on years outside a chart's range. (5) A pre-registered repair experiment in which cropping

[1] Code and results: https://github.com/the-neemon/type-aware-vlm-failures

rescues misreads more than fabrications in both models, and a controller evaluated against selective prediction.

2 Related Work

Pre-generation failure detection. Models often know when they are likely to be wrong: language models' own estimates of correctness are informative (Kadavath et al., 2022), and hallucination risk in VLMs can be read from internal representations in a single forward pass, at up to 0.93 AUROC across eight models (Kogilathota et al., 2026). HALP also finds that the most informative features differ by architecture, with Qwen2.5-VL-7B best served by visual-only features. Zhang et al. (2026) isolate hallucination-relevant attention heads with a variational information bottleneck and steer with probe gradients. Both address binary object hallucination on photographs; neither asks whether different kinds of failure are internally distinguishable.

Failure types inside the model. Liu et al. (2026) find, by activation patching across five VLMs, a visual grounding pathway behind correct predictions and a separate pathway driving errors, at different depths; the circuit transfers to relational but not attribute hallucination. This selectivity is the closest prior evidence that hallucination is not one phenomenon, and it motivates probing every layer.

Chart reasoning and error taxonomies. ChartQA (Masry et al., 2022) is the standard chart question answering benchmark. RADAR (Rani et al., 2025) and DRAGON (Iyengar et al., 2026) annotate the regions that justify an answer; they label *where* a model should look, not *how* it failed. Ashury-Tahan et al. (2026) build failure taxonomies for text-only LLMs with an LLM judge; we adopt the method, not the taxonomy. Visual Self-Refine (Li et al., 2026) makes a chart parser re-examine its own localisations. Re-looking is thus established; whether the right repair depends on the failure type is not.

Probes and selective prediction. Linear probes read information from intermediate layers (Alain and Bengio, 2016), but a probe can succeed by exploiting properties of the input rather than of the model, so it needs controls (Hewitt and Liang, 2019). Sahoo et al. (2026) show reasoning type probes at 100% accuracy collapsing to chance once format confounds are residualised out, and Yuan et al. (2026) find a hidden-state error probe at 0.95 AUROC whose signal supported no successful intervention. We therefore report every probe beside an activation-free baseline and test repairs directly. We evaluate abstention with risk-coverage curves, as in selective classification (Geifman and El-Yaniv, 2017).

3 Experimental Setup

Models and inference. We use two frozen VLMs, Qwen2.5-VL-7B (Bai et al., 2025) and LLaVA-NeXT with a Mistral-7B language model (Liu et al., 2024). Each question gets the standard ChartQA suffix "Answer the question using a single word or phrase." and is decoded greedily (at most 32 tokens) in fp16 on RTX 2080 Ti GPUs. Answers are scored by ChartQA relaxed accuracy: numbers within 5% of the gold, strings by exact match.

Activations. Forward hooks record the residual stream after every decoder layer during the prefill of the same `generate()` call that produces the answer, so each vector belongs to the scored answer. We pool four positions per layer (mean and max over image tokens, the last prompt token, and the mean over prompt text tokens) for all 28 layers of Qwen ($d = 3584$) and 32 of LLaVA ($d = 4096$). The image precedes the question and attention is causal, so image-token states cannot depend on the question.

Probes. A probe is an L_2-regularised logistic regression on one pooled vector, z-scored with training-set statistics, with λ chosen from $\{1, 10, 10^2, 10^3, 10^4\}$. The binary probe uses a 60/20/20 split by *figure*: layer, position and λ are chosen on validation and the test set is scored once. Type probes, whose classes are smaller, use five folds by figure: within each outer fold, layer and λ are chosen by inner cross-validation on the other folds, so every item is scored by a probe that never saw its chart; the position is fixed to the last prompt token. We report AUROC with 95% intervals from a bootstrap over figures (2,000 draws), since questions on one chart are correlated. Seed 42 throughout.

Baselines and controls. Every probe is reported beside a baseline without activations. On ChartQA this is the same probe on four surface features (question source, question length, number of image tokens, numeric gold). We also *residualise*: regress

Label	Qwen n	Qwen %	LLaVA n	LLaVA %
STRUCTURAL	71	22	583	49
COMPUTATION	93	29	435	37
NOT_AN_ERROR	112	35	69	6
AMBIGUOUS	40	13	74	6
FABRICATION	1	0.3	17	1.4
errors	317		1,178	

Table 1: Labels of each model's ChartQA test errors.

	Probe	5 splits	Resid.	Surface
Qwen, all	0.877 [.83,.92]	.862	.803	.715
Qwen, clean	0.932 [.90,.96]	.910	.901	.786
LLaVA, all	0.798 [.76,.84]	.801	.783	.608
LLaVA, clean	0.819 [.78,.85]	.810	.798	.636

Table 2: E1 binary probe (wrong vs. right), test AUROC [95% CI]; mean over five figure splits; with surface features regressed out; and the surface baseline. Clean: without NOT_AN_ERROR items.

the surface features out of the activations, fitted on training rows only, and retrain (Sahoo et al., 2026). On synthetic charts the baselines are a question-text classifier, the image-token probe, and the probe on chart metadata (pilot, axis range, tick density, number of bars, lookalike name, phrasing).

4 Labelling the ChartQA Errors

On the 2,500 ChartQA test questions (1,250 human-written, 1,250 machine-augmented, 1,509 charts), Qwen is correct on 87.3% and LLaVA on 52.9%, leaving 317 and 1,178 wrong answers. Every wrong answer was labelled with five classes: STRUCTURAL (explained by misreading the figure), FABRICATION (content the figure does not support), COMPUTATION (inputs read correctly, arithmetic or logic wrong), NOT_AN_ERROR (actually right: an equivalent format, a wrong gold answer, or a valid reading of the question) and AMBIGUOUS. An LLM annotator (Claude) labelled all errors from the chart, question, gold and model answer, never seeing the model's identity, following a written rubric (Appendix A gives examples). Before labelling LLaVA it re-labelled 50 random Qwen errors blind and agreed on 47 ($\kappa = 0.917$).

Table 1 gives three findings. **Fabrication is almost absent from ChartQA** (1 of 317 and 17 of 1,178 errors): its questions are written about content the chart contains, so a model rarely has occasion to invent, and most apparent inventions turned out to be garbled printed values (486 for a printed 4,863), which are misreads. **35% of Qwen's scored errors are not errors** (6% of LLaVA's), mostly equivalent formats (0.95 for 95%) and wrong gold answers. **The models fail differently**: half of LLaVA's errors are misreads, while Qwen's real errors are mostly arithmetic. The naturalistic arm therefore supports a misread versus computation contrast, and fabrication needs the synthetic arm (Section 6).

Human agreement. Two team members labelled the same 200 errors blind (100 per model, stratified by the LLM label). They agreed with each other at $\kappa = 0.421$ (Qwen 0.452, LLaVA 0.345), below our 0.6 gate. After they adjudicated the 79 disagreements together, the human labels agree with the LLM labels at $\kappa = 0.614$ overall (Qwen 0.589, LLaVA 0.613; 72% raw agreement). All experiments use the LLM labels.

5 Real Charts: Predicting Errors and Their Type

Will the answer be wrong? (E1, E2). A sanity probe first confirmed that each cache decodes what it plainly contains (human-written vs. augmented question: validation AUROC 1.000 for Qwen, 0.999 for LLaVA). The binary probe then predicts wrong answers far above the surface baseline in both models (Table 2), always at the last prompt token (Qwen layer 21 of 28, LLaVA 24 of 32); image-token positions stay near the surface baseline at every depth (Appendix B). Removing the NOT_AN_ERROR items raises Qwen's probe from 0.877 to 0.932: part of its apparent error was label noise. The signal is not question style. With the surface features regressed out, the probe loses only 0.02 to 0.07 and stays well above the surface features alone (Table 2), so in both models it reads an internal property of the hidden state beyond what the question's surface reveals. It also holds within human-written questions only (0.895 Qwen, clean; 0.850 LLaVA) and moves by about ± 0.02 across five figure splits. Applied unchanged to 960 human-written validation questions on charts never used in training, Qwen's probe scores 0.772 [0.73, 0.81]. Its probabilities are calibrated (expected calibration error 0.03 for Qwen, 0.05 for LLaVA).

Misreads vs. arithmetic errors (E3). ChartQA supplies enough misreads and computation errors, but not fabrications, to train type probes (Table 3).

Probe (trained on)	Qwen	LLaVA
Misread vs. correct	**0.892** [.85,.93]	**0.813** [.79,.83]
surface	0.651	0.536
retrieval qs. only	0.913	0.805
residualised	0.830	0.798
on comp. vs. corr.	0.820	0.586
Comp. vs. correct	**0.904** [.87,.93]	**0.949** [.94,.96]
surface	0.818	0.821
arithmetic qs. only	0.845	0.836
residualised	0.874	0.900
on misread vs. corr.	0.732	0.540
Misread vs. comp.	**0.897** [.84,.94]	**0.961** [.95,.97]
surface	0.693	0.845
residualised	0.918	0.920
errors (misread / comp.)	71 / 93	583 / 435

Table 3: E3 on ChartQA (comp.: computation): out-of-fold AUROC [95% CI] of each probe on its own contrast, its surface baseline, the probe trained within one question type, with surface features regressed out, and applied to the other error type.

		Qwen	LLaVA
Shown bar, value	read right	554/600	185/600
Missing, value	rejected	15	0
	"0"	494	232
	fabricated	91	368
Missing, compare	fabricated	37	101
	picked shown	263	199
Missing, neighbor	fabricated	300	290
Missing-vs-shown probe, test AUROC			
value		0.999	0.977
compare		0.991	0.959
neighbor		1.000	0.996
text-only baseline		0.49–0.52	0.49–0.52
image-token probe		0.500	0.500

Table 4: Pilots 1 and 2. Top: answers (value rows pool both pilots; compare and neighbor are pilot 2, 300 each). Bottom: probes separating a missing from a shown name under identical wording (pilot 2, held-out charts).

Misreads are detectable well above the surface features (0.892 and 0.813), and misreads and computation errors are separable (0.897 and 0.961). In LLaVA each type probe barely responds to the other type (0.586 and 0.540); Qwen's misread probe also flags computation errors (0.820), but surface features alone reach 0.765 on that contrast. Computation errors only occur on questions that ask for an operation, and that wording is visible from the first layer (LLaVA's computation probe already scores 0.865 at layer 0). We therefore tagged each question as *arithmetic* or *retrieval* with a keyword rule fixed in advance and retrained within one type: the computation probe still reaches 0.845 and 0.836 within arithmetic questions, and the misread probe 0.913 and 0.805 within retrieval questions. Residualised, all three probes stay between 0.80 and 0.92 in both models. On real charts, then, the two error types the models actually make leave different, linearly readable signatures.

6 Synthetic Charts: Inducing Fabrication

Design. We induce fabrication by asking about something the chart does not show. Two constraints shape the design: a probe comparing ChartQA misreads with synthetic fabrications would learn "real vs. synthetic chart", and on synthetic data the question template alone predicts the intended failure type perfectly. Every contrast is therefore made *within one question wording*, and every outcome is scored from the model's actual answer. **Pilot 1** (300 bar charts, 4 to 7 bars, six themes) asks each chart the same question twice: once about a shown bar ("What is the value of Mango?") and once about a plausible missing name ("... of Guava?"). **Pilot 2** (300 charts) adds lookalike names (asked for Dubai when Dublin is shown), values off multiples of 5 on axes up to 200, and two more question families: *compare* ("Which is larger, Guava or Apple?") and *neighbor* ("Which bar is right of Guava?"). **Pilot 3** (300 charts of consecutive years) asks about a shown year and, in the same words, about a year one to three years outside the shown range. A missing-bar answer is scored as *rejected* ("not shown"), *"0"* (no bar has zero height, so we treat it as a refusal expressed as a number, and score it wrong but not as fabrication), *fabricated* (any other number, or naming a bar), or *picked shown* (naming the shown bar in a compare question, reported separately).

The models know the bar is missing, and answer anyway. Neither model refuses: Qwen rejects 15 of 1,200 missing-name questions and LLaVA none (Table 4). Qwen mostly answers "0", and when it invents a number it usually copies the lookalike bar (57 of its 76 pilot-2 inventions); LLaVA invents far more often (368 vs. 91). Yet probes on the last prompt token separate missing from shown names almost perfectly, also for lookalike names (0.98 to 1.00 Qwen, 0.92 to 0.99 LLaVA). Two controls rule out shortcuts: a classifier on the question text is at chance, because every name is shown on some charts and missing on others, and the image-token probe scores exactly 0.500, because image tokens precede the question and cannot see it. The signal

Probe	Qwen S/C	Qwen F/C	Qwen F/Z	LLaVA S/C	LLaVA F/C	LLaVA F/Z
Misread (S/C)	**.887**	.594	.400	**.776**	.879	.215
metadata	.814	.557	.543	.512	.522	.537
+ gridline†	.920			.860		
Fabricated (F/C)	.576	**1.00**	.286	.680	**.988**	.243
Invented vs. "0"	.336	.138	**.984**	.520	.226	**.986**

Table 5: E3 cross-transfer on synthetic value questions (pilots 1 and 2, 5-fold CV by chart). Rows: the probe; columns: the contrast it is scored on (C correct read, S misread, F invented number, Z "0"). Bold: own task. Metadata: the same probe on chart settings. †Gridline features chosen after seeing the data.

	Qwen	LLaVA
Answer to the out-of-range year		
Refused	119	0
"0"	76	19
Made-up number	105	281
Shown year read correctly	273	74
Missing-vs-shown probe		
layer	L10	L18
out-of-range vs. shown	1.000 [1.00, 1.00]	0.911 [0.89, 0.93]
made-up vs. correct	1.000	0.922
made-up vs. refused	0.521	–
misread vs. correct	0.494	0.568
Fabrication probe (E3)		
layer	L10	L19
out-of-range vs. shown	1.000 [1.00, 1.00]	0.880 [0.85, 0.90]

Table 6: Pilot 3: each chart is asked about a shown year and, in the same words, about a year outside the shown range. Top: what the models answered to the out-of-range year (300 per model). Bottom: AUROC of the two missing-bar probes, trained on pilots 1 and 2 only and applied unchanged, with 95% figure-bootstrap CIs; layer chosen on pilots 1 and 2. Qwen's refusals include 17 written as sentences that name a year, which the scorer missed and we counted by hand; the made-up vs. refused AUROC uses the 102 refusals the scorer recognised.

appears early (Qwen by layer 2, LLaVA by about layer 9).

Misreads vs. fabrications (E3). On the value template the four classes are correct read, misread, invented number and "0" (Qwen 554, 46, 91, 494; LLaVA 185, 415, 368, 232). In Qwen the two type probes are largely specific (Table 5): the misread probe barely responds to fabrications (0.594), and the fabrication probe barely to misreads (0.576). In LLaVA the misread probe fires *more* on fabrications (0.879) than on its own task (0.776), the pattern our proposal called one signal wearing two labels. In both models the "fabrication" probe is a missing-bar detector: it ranks "0" answers as high as invented numbers. What separates an invented number from "0" is decodable (0.984 and 0.986), but only in the last layers, where it reads out the answer about to be produced (Appendix B).

A chart property explains synthetic misreads. Relaxed accuracy tolerates a 5% *relative* error, so whether a coarse reading is a misread depends on how far the true value lies from a gridline relative to its size. Two features declared in advance (distance to the nearest gridline, and to the nearest multiple of 10) added to the metadata reach 0.853 (Qwen) and 0.642 (LLaVA), below both probes; the same distances divided by the value, added after seeing that result, reach 0.920 and 0.860, above both. The synthetic misread signal may describe the chart more than the model.

The missing-bar signal is about answerability (pilot 3). In pilots 1 and 2 a fabrication always concerns a missing category. Pilot 3 separates the two (Table 6). Here Qwen refuses 119 of 300 out-of-range years, against 15 of 1,200 missing names, while LLaVA still invents a number for 281. Most invented numbers copy a shown bar's value (92 of 105 for Qwen, 135 of 281 for LLaVA) rather than extend the trend (5 and 9). Both missing-bar probes, trained only on missing *names* in pilots 1 and 2, flag out-of-range *years* perfectly in Qwen and at 0.911 and 0.880 in LLaVA, are at chance in the first layers, and ignore misreads of shown years. In Qwen the probe fires equally whether the model then refuses, says "0" or invents a number (invented vs. refused 0.521): it encodes that the chart cannot answer the question, not what the model will do about it.

7 Does the failure type decide the repair? (E4)

We re-query all erroneous value questions in synthetic pilots 1 and 2 (631 for Qwen, 1,015 for LLaVA) to test whether the best repair depends on failure type. Types come from the original answers, independently of probes: misreads of shown bars (S), invented nonzero values for missing bars (F), and missing-bar "0" answers (Z). We score Z as wrong, separately from fabrication.

We compare four repairs: a sampled re-ask of the original question (temperature 0.7); whole-chart upsampling; a question-conditioned crop retaining the target bar and y-axis strip; and a verification

Original answer	n	Re-ask	Upsample	Crop	Verify
Qwen2.5-VL-7B					
Misread	46	0.11	0.13	**0.26**	0.09
Made-up number	91	0.00	0.00	0.00	**0.14**
"0"	494	0.02	0.02	0.02	**0.69**
crop DiD			+0.261 [+0.140, +0.389]		
LLaVA-NeXT-7B					
Misread	415	0.03	0.01	**0.15**	0.03
Made-up number	368	0.00	0.00	0.00	0.00
"0"	232	0.00	0.00	0.00	0.00
crop DiD			+0.152 [+0.119, +0.189]		

Table 7: E4 recovery rates on initially wrong value answers. Bold marks each row's best repair, excluding zero ties. DiD compares crop's recovery gain on misreads with its gain on made-up numbers. Brackets give 95% CIs from figure bootstraps; missing-bar "0" answers form a separate row.

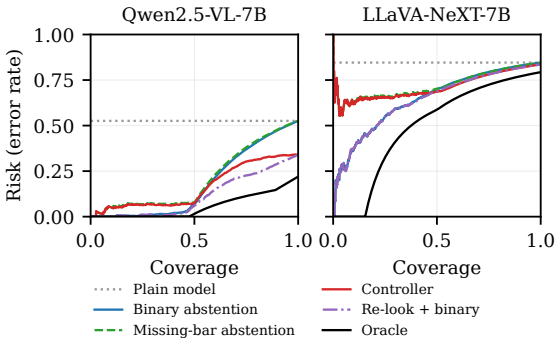


Figure 1: Risk versus coverage on synthetic value questions. The controller and Re-look + binary use the same repaired answers with different abstention rankings. At full coverage, the reduction in risk is the net gain from re-looking. Curves average over tied confidence scores.

prompt requesting chart evidence before the final answer. The latter three use greedy decoding. Upsample and crop share a 1,000,000-pixel budget. Recovery requires a correct read under the original 5% numeric tolerance, or an explicit refusal for a missing bar. The re-ask controls for improvement from another attempt.

Before running the interventions, we pre-registered the difference in recovery gains between S and F as the primary contrast:

$$\Delta_{\text{crop}} = (r_{S,\text{crop}} - r_{S,0}) - (r_{F,\text{crop}} - r_{F,0}), \quad (1)$$

where r denotes recovery rate. Baseline recovery is zero because we select only errors; the contrast excludes Z. We use 2,000 figure-bootstrap resamples for 95% confidence intervals (CIs).

The crop DiDs are +0.261 for Qwen and +0.152 for LLaVA, with CIs excluding zero as predicted (Table 7). The pre-registered secondary verify DiD is -0.056 [-0.162, +0.053] for Qwen, including zero, and +0.031 [+0.015, +0.050] for LLaVA.

Qwen's best repair is crop for misreads and verify for fabrications and Z answers, recovering 14.3% and 69.2% respectively. Misreads and fabrications select different best repairs in 99% of bootstrap resamples. LLaVA's crop helps misreads, but no repair recovers any of its 600 missing-bar errors. All repairs tie on these errors, making an argmax change uninformative. Re-asking turns 215 of its 232 Z answers into invented numbers.

Crop falls back to upsampling when the bar is missing. The two repairs give identical answers on all 1,185 missing-bar errors across models, so the primary contrast cannot isolate localisation. Comparing crop with upsample isolates this gain: +0.145 [0.111, 0.181] for LLaVA and +0.130 [-0.023, 0.289] for Qwen. Qwen's interval includes zero, with only 46 misreads available.

8 A type-aware controller (E5)

We compare a type-aware controller with binary selective prediction on all 1,200 value questions per model, including correct reads and refusals. Three linear probes on the last prompt-token hidden state score wrong versus right, missing versus shown bar, and misread versus correct read before generation. Scores use five figure folds: each probe trains on the other four, selecting layer and regularisation within its training folds.

The *controller* re-looks above a misread-score threshold and withholds questions with the highest missing-bar scores first. We choose the repair and threshold to maximise accuracy on four folds, then apply them to the fifth. Baselines include the plain model, binary abstention, missing-bar abstention, and *Re-look + binary*, which uses the controller's re-look rule with binary ranking. The label-aware oracle chooses the best action for each true class and withholds the least recoverable classes first. Re-query outcomes on initially correct answers count harmful repairs against each policy: crop breaks 133 of Qwen's 554 correct reads.

Coverage is the fraction retained; risk is their error rate. AURC averages risk over coverage (lower is better), averaging tied confidence scores. Correct refusals count as correct answers; policy abstention removes questions from coverage. CIs use 1,000 figure-bootstrap resamples.

Adding a re-look lowers AURC under either ab-

Policy	AURC	95% CI
Qwen2.5-VL-7B		
Plain model	0.526	[0.513, 0.538]
Binary abstention	0.179	[0.169, 0.190]
Missing-bar abstention	0.208	[0.191, 0.226]
Controller	0.162	[0.145, 0.180]
Re-look + binary	0.114	[0.103, 0.126]
Oracle	0.056	[0.049, 0.064]
re-look gain (binary)	−0.065	[−0.073, −0.056]
re-look gain (missing-bar)	−0.045	[−0.054, −0.037]
LLaVA-NeXT-7B		
Plain model	0.846	[0.827, 0.864]
Binary abstention	0.645	[0.608, 0.681]
Missing-bar abstention	0.725	[0.686, 0.759]
Controller	0.715	[0.674, 0.750]
Re-look + binary	0.638	[0.600, 0.674]
Oracle	0.497	[0.464, 0.528]
re-look gain (binary)	−0.007	[−0.012, −0.003]
re-look gain (missing-bar)	−0.010	[−0.017, −0.005]

Table 8: E5 AURC on 1,200 value questions per model, with 95% figure-bootstrap CIs. Each re-look gain compares policies with the same abstention ranking: re-look plus binary minus binary, or controller minus missing-bar abstention. A negative difference means re-looking helps. The oracle uses true classes.

stention ranking in both models (Table 8, Figure 1; all four gain CIs exclude zero). The controller beats binary abstention only for Qwen: AURC difference -0.017 $[-0.028, -0.004]$, compared with +0.069 [0.045, 0.094] for LLaVA. Without re-looking, binary ranking also outperforms missing-bar ranking because it flags misreads as well as missing-bar errors. Re-look + binary has the lowest AURC of the evaluated probe-based policies in both models (0.114 and 0.638).

Qwen selects verify in every fold. Of 222 fixes, 212 are Z answers converted to refusals, seven are fabrications, and only three are misreads; one correct refusal becomes wrong. Most gains are refusals on missing-bar questions, although the re-look was intended to correct misreads. The misread probe also responds to missing bars and therefore triggers this repair. LLaVA selects crop in every fold. It fixes 12 misreads and re-looks at 437 missing-bar questions without recovering any. With access to true classes, the oracle achieves lower AURCs of 0.056 and 0.497, so better policies are possible within this evaluation.

9 Discussion

E1 predicts wrong answers (AUROC 0.877 for Qwen, 0.798 for LLaVA; surface baselines 0.715 and 0.608). Real-chart probes distinguish misreads from computation errors (0.897 and 0.961), including after surface residualisation (0.918 and 0.920). This partly supports type separability for a different pair from the proposed misread/fabrication contrast. Question wording may still explain part of the separation: the surface controls do not remove question type.

The synthetic results depend on the model. Qwen's misread probe responds more strongly to misreads (0.887) than fabrications (0.594); LLaVA's responds more to fabrications (0.879) than misreads (0.776). Post-hoc gridline features with chart metadata exceed both probes on misreads (0.920 and 0.860). We therefore cannot infer a general mechanism shared by the models from separability alone.

In pilot 3, the missing-versus-shown probes trained on absent category names detect out-of-range years (AUROC 1.000 and 0.911). Qwen's probe barely separates fabrication from correct refusal (0.521): its score tracks whether the chart supports answering, with little separation between these responses. Qwen refuses 119 of 300 out-of-range questions, compared with 15 of 600 absent-name questions. Its response differs between kinds of unsupported questions; chart design also changes between these pilots.

Repair effects depend on failure type in E4, yet the type-aware controller in E5 improves on binary abstention only for Qwen, mainly by prompting refusals. For LLaVA's missing-bar errors, policy abstention is the only effective option we tested. These results favour using answerability together with a general error score, and assessing the repair and abstention ranking as parts of the same policy.

10 Conclusion

Before a VLM answers a chart question, a linear probe on its last prompt token predicts whether the answer will be wrong, beyond what the question's surface reveals. On real charts the errors are misreads and arithmetic errors, and the two leave different internal signatures in both models. Fabrication is rare on real charts but easy to induce: both models encode that a question asks for something the chart does not show, and answer anyway. That answerability signal, together with a general error score, is the most useful input for deciding when to abstain; whether to re-look depends on the failure type, as the pre-registered crop test shows, but in our controller re-looking helped mainly by turning missing-bar answers into refusals.

Limitations, Challenges and Future Work

Labels. The ChartQA labels come from one LLM annotator. Two human annotators agreed with each other below our 0.6 gate, so the five-way taxonomy is hard to apply consistently, and the adjudicated human labels match the LLM labels only moderately. Type probes trained on these labels inherit their noise.

Scope of the type contrast. Fabrication is nearly absent from ChartQA, so the misread versus fabrication contrast rests on synthetic bar charts with a single value-question template, where a fabrication always concerns something missing. Synthetic misreads are partly determined by where a value sits relative to the gridlines, and the best gridline features were chosen after seeing the data. Qwen makes few misreads (46 synthetic, 71 real), so its intervals are wide. The question-type rule is a keyword heuristic.

Repairs and the controller. E4 and E5 use synthetic charts only, because the crop needs bar geometry that ChartQA does not provide. Their balance of shown and missing bars differs from how often unsupported questions occur in real use, so the controller's risk-coverage numbers should not be extrapolated to ChartQA. Cross-fitting by figure still reuses the same pilots, and we did not optimise latency or inference cost. All probes are linear and diagnostic; we did not test whether steering along the missing-bar direction changes the answer.

Future work. Repair selection on naturalistic unanswerable questions; a crop for real charts based on text detection; more models and chart types, including node-link diagrams; and a causal test of the missing-bar direction.

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A Examples of the error labels

Examples from Qwen's labels (question; gold / model answer; the annotator's rationale, paraphrased). STRUCTURAL: "How many Swedish companies operated on the total market … 2018?"; 315 / 347; 347 is the adjacent 2017 bar. COMPUTATION (arithmetic error): "What is the average percentage in Live event category?"; 35.2 / 38.2; the printed values 35, 53, 38, 30 and 20 average 35.2, so the model added wrongly. NOT_AN_ERROR: "What is the share of one-year olds vaccinated against HepB3 in Bhutan?"; 95 / 0.95; the same share as a fraction, scored wrong only for its format. AMBIGUOUS: "How many colors are used in the bar?"; 11 / 10; two near-identical teal shades make 10 or 11 depend on telling them apart. FABRICATION (Qwen's only case): a pie chart with slices 35, 40, 13 and 11; 75 / 55; 55 matches no slice and no sum of slices.

B Layer profiles

Figure 2 shows the binary probe by relative depth: the last prompt token rises from about 0.70 at the first layer to a plateau over the last third of each network (best at layer 21 of 28 for Qwen, 24 of 32 for LLaVA), while image-token positions stay at 0.62 to 0.68, near the surface baseline. For chart errors the informative state is the one that has read the question, unlike HALP's finding for object hallucination (Kogilathota et al., 2026). Figure 3 shows the synthetic type probes: the missing-bar signal saturates early (Qwen by layer 3, LLaVA by about layer 8), the misread signal builds gradually and peaks in the last third.

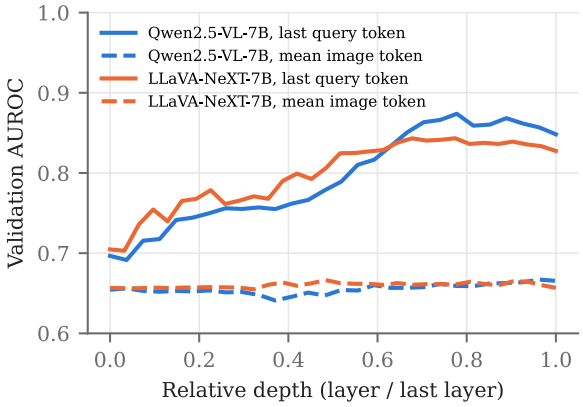


Figure 2: E1 binary probe, validation AUROC by relative depth. Solid: last prompt token; dashed: mean over image tokens.

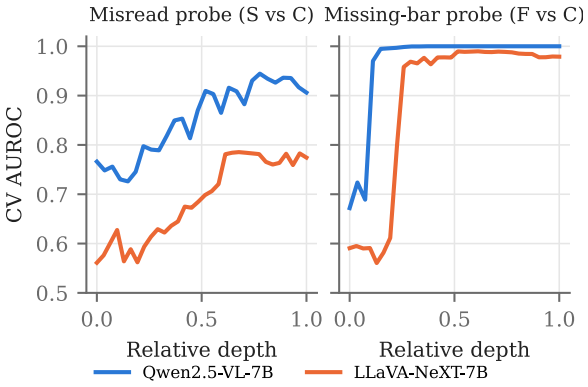


Figure 3: Synthetic type probes, out-of-fold AUROC on their own task by relative depth.

C Reproducibility

All runs used RTX 2080 Ti GPUs in fp16 with SDPA attention, PyTorch 2.8.0, Transformers 4.57.6 and seed 42, with greedy decoding of at most 32 new tokens; the full environment is pinned in `requirements.txt`. Synthetic charts are rendered with Matplotlib 3.10.0 and are deterministic given the seed; the SHA-256 hashes of the three manifests (pilots 1 to 3) begin `20e80119`, `41fc27e5` and `ed311841`. Activations are pooled in fp16 and kept off the repository. Every number in this report is read from a file in `results/`, and the repository README lists the command that produces each file.